Year 13 Maths - Similar to June 2024 Question 1

Similar questions - Pure & Mechanics

Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 In this question $x > 0$.

(a) Expand $\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^2$. **[2]**

(b) Hence show that $\left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right)^2 - \left(x^{\frac{1}{2}} - x^{-\frac{1}{2}}\right)^2 = 4$. **[2]**

(c) Given that $x^{\frac{1}{2}} + x^{-\frac{1}{2}} = 3$, find the exact value of $x + \dfrac{1}{x}$ and the exact values of $x^{\frac{1}{2}} - x^{-\frac{1}{2}}$. **[3]**

2 A rectangle has area $(2x^2 + 5x - 12)$ cm^2 and length $(x + 4)$ cm.

(a) Show that the width of the rectangle is $(2x - 3)$ cm. **[2]**

(b) Given that the perimeter is 26 cm, find the value of x. **[2]**

(c) Find the exact length of a diagonal of the rectangle when x takes this value. **[2]**

Question 1

(a) $x+2x^{\frac{1}{2}}x^{-\frac{1}{2}}+x^{-1}=x+2+\dfrac{1}{x}$

(b) $\left(x+2+\dfrac{1}{x}\right)-\left(x-2+\dfrac{1}{x}\right)=4$

(c) $x+2+\dfrac{1}{x}=9\Rightarrow x+\dfrac{1}{x}=7$. Then $\left(x^{\frac{1}{2}}-x^{-\frac{1}{2}}\right)^2=x-2+\dfrac{1}{x}=5$, so $x^{\frac{1}{2}}-x^{-\frac{1}{2}}=\pm\sqrt{5}$

Question 2

(a) $2x^2+5x-12=(2x-3)(x+4)$, so width $=\dfrac{(2x-3)(x+4)}{x+4}=2x-3$

(b) $2\left[(x+4)+(2x-3)\right]=26\Rightarrow 3x+1=13\Rightarrow x=4$

(c) Length $=8$, width $=5$, so diagonal $=\sqrt{64+25}=\sqrt{89}$ cm